Power Rule Discoveries

Let's see if we can discover an easy and very useful rule for finding the ***derivatives of power functions.***

REMEMBER: Derivatives are slopes of curves at one point.

Quadratic

Examine the data table below. In the first column are *x*-values; in the second column are *y*-values for the function $f(x) = x^2$; in the third column are slope values for that function.

x	*f(x)*	*slopes*
-5	25	-10
-3.5	12.25	-7
-1.25	1.5625	-2.5
0	0	0
2	4	4
4.75	22.5625	9.5
7	49	14

Compare the slopes for the function with the *x*-values.

- Do you see a pattern? Yes.
- What is the pattern?

The slope of x^2 at any point is 2 times the x-value.

Cubic

Examine the data table below. In the first column are *x*-values; in the second column are *y*-values for the function $f(x) = x^3$; in the third column are slope values for that function.

x	*f(x)*	*slopes*
-5	-125	75 ÷ 3 = 25
-3	-27	27 ÷ 3 = 9
-1	-1	3 ÷ 3 = 1
0	0	0 ÷ 3 = 0
2	8	12 ÷ 3 = 4
4	64	48 ÷ 3 = 16
6	216	108 ÷ 3 = 36

Compare the slopes for the function with the *x*-values.

- Do you see a pattern? no
- What is the pattern?

HINT: This pattern is not so easy to see. Try dividing each of the slopes in the third column by 3.

- Now, do you see a pattern? yes
- What is the pattern? The new slopes are the x-value squared.
- Describe how the numbers in the slope column are related to the *x*-values.

The slopes of x^3 are 3 times x^2.

Quartic

Examine the data table below. In the first column are *x*-values; in the second column are *y*-values for the function $f(x) = x^4$; in the third column are slope values for that function.

x	*f(x)*	*slopes*	
-5	625	-500 ÷ 4	= -125
-3	81	-108 ÷ 4	= -27
-1	1	-4 ÷ 4	= -1
0	0	0 ÷ 4	= 0
2	16	32 ÷ 4	= 8
4	256	256 ÷ 4	= 64
6	1296	864 ÷ 4	= 216

Compare the slopes for the function with the *x*-values.

- Do you see a pattern? no
- What is the pattern?

HINT: This pattern is not so easy to see either. Try dividing each of the slopes in the third column by 4.

- Do you see a pattern? yes
- What is the pattern? the new slopes are the x-value cubed.
- Describe how the numbers in the slope column are related to the *x*-values.
  The slopes of x^4 are 4 times x^3.

Quintic

Examine the data table below. In the first column are *x*-values; in the second column are *y*-values for the function $f(x) = x^5$; in the third column are slope values for that function.

x	*f(x)*	*slopes*	
-5	-3125	3125 ÷ 5	= 625
-3	-243	405 ÷ 5	= 81
-1	-1	5 ÷ 5	= 1
0	0	0 ÷ 5	= 0
2	32	80 ÷ 5	= 16
4	1024	1280 ÷ 5	= 256
6	7776	6480 ÷ 5	= 1296

Compare the slopes for the function with the *x*-values.

- Do you see a pattern? no, but I bet if I divide by 5 I will see a pattern.
- What is the pattern?
  the new slopes are the x-value to the 4th power.
- Describe how the numbers in the slope column are related to the *x*-values.
  The slopes are 5 times x^4.

What do you think the rule for the slopes of the function, $f(x) = x^n$ is?

n times x^{n-1}

State the rule for finding the derivative of a power function in words.

The derivative of a power function, x^n, is the power times the x to the one less power.

Click on the link in ILearn for the **Derivative App**. To use the app, choose the **quadratic function** and read the directions above the graph. Position the graph so that you can see the whole thing and go to work on the green dots.

When you have correctly placed all the green dots, you will see a graph and formula for the slopes of the function. **Does it match your expectation?** Yes

Do the **cubic**, **quartic**, and **quintic** functions in the app as well, but don't do any of the others yet. Do your results match your expectations? yes

Does this rule work for other kinds of exponents besides just positive whole numbers? I don't know.

Let's look at some more complicated power functions.

Negative 1 Power

Examine the data table below. In the first column are *x*-values; in the second column are *y*-values for the function $f(x)=\frac{1}{x}$ *or* $f(x)=x^{-1}$; in the third column are slope values for that function.

x	*f(x)*	*slopes*
-1/2	-2	-4
-1/3	-3	-9
-1	-1	-1
0	#DIV/0!	#DIV/0!
2	1/2	-1/4
4	1/4	-1/16
6	1/6	-1/36

Compare the slopes for the function with the *x*-values.

- Do you see a pattern? Yes
- What is the pattern? the slopes are negative and 1 over the x-value square.
- Describe how the numbers in the slope column are related to the *x*-values.

the slopes are $-\frac{1}{x^2}$.

Negative 2 Power

Examine the data table below. In the first column are *x*-values; in the second column are *y*-values for the function $f(x)=\frac{1}{x^2}$ *or* $f(x)=x^{-2}$; in the third column are slope values for that function.

x	*f(x)*	*slopes*
-1/2	4	16 ÷ -2 = -8
-1/3	9	54 ÷ -2 = -27
-1	1	-2 ÷ -2 = 1
0	#DIV/0!	#DIV/0!
2	1/4	-1/4 ÷ -2 = $\frac{1}{8}$
4	1/16	-1/32 ÷ -2 = $\frac{1}{64}$
6	1/36	-1/108 ÷ -2 = $\frac{1}{216}$

Compare the slopes for the function with the *x*-values.

- Do you see a pattern? not really. I will try dividing by the power -2.
- What is the pattern? the new slopes are the same sign as x and are 1 over the x-value cubed.
- Describe how the numbers in the slope column are related to the *x*-values.

the slopes are -2 times $\frac{1}{x^3}$.

Does it look like the rule works for negative slopes? Do you suppose it works for fractional slopes?

Yes

thinking, maybe yes.

Fractional Power – Square Root

Examine the data table below. In the first column are *x*-values; in the second column are *y*-values for the function $f(x)=\sqrt{x}$ *or* $f(x)=x^{1/2}$; in the third column are slope values for that function.

x	*f(x)*	*slopes*	
0	0	#DIV/0!	
1	1	1/2 ÷ ½	= 1
4	2	1/4 ÷ ½	= ½
9	3	1/6 ÷ ½	= ⅓
16	4	1/8 ÷ ½	= ¼
25	5	1/10 ÷ ½	= ⅕
36	6	1/12 ÷ ½	= 1/6

Compare the slopes for the function with the *x*-values.

- Do you see a pattern? not really. I will try dividing by the power, ½.
- What is the pattern? the new slopes equal to 1 over the y values
- Describe how the numbers in the slope column are related to the *x*-values.

The slope are ½ times 1/y

$= \frac{1}{2} \cdot \frac{1}{\sqrt{x}}$.

Fractional Power – Cube Root

Examine the data table below. In the first column are *x*-values; in the second column are *y*-values for the function $f(x)=\sqrt[3]{x}$ *or* $f(x)=x^{1/3}$; in the third column are slope values for that function. Compare the slopes for the function with the *x*-values. Do you see a pattern? What is the pattern? Describe how the numbers in the slope column are related to the *x*-values.

x	*f(x)*	*slopes*	
-27	-3	1/27 ÷ ⅓	= 1/9
-8	-2	1/12 ÷ ⅓	= ¼
-1	-1	1/3 ÷ ⅓	= 1
0	0	#DIV/0!	
1	1	1/3 ÷ ⅓	= 1
8	2	1/12 ÷ ⅓	= 4
27	3	1/27 ÷ ⅓	= 1/9

Compare the slopes for the function with the *x*-values.

- Do you see a pattern? no. I will try dividing by the power ⅓.
- What is the pattern? the new slopes are equal to 1 over the y-value squared.
- Describe how the numbers in the slope column are related to the *x*-values.

The slopes are ⅓ times $1/y^2$

$= \frac{1}{3} \cdot \frac{1}{(x^{1/3})^2} = \frac{1}{3} \cdot \frac{1}{x^{2/3}}$

Does it look like the rule works for fractional powers? yes

Click on the link in ILearn for the **Derivative App**. To use the app, choose the **rational 1 function** and read the directions above the graph. Position the graph so that you can see the whole thing and go to work on the green dots.

When you have correctly placed all the green dots, you will see a graph and formula for the slopes of the function. **Does it match your expectation?** yes

Do the **rational 2**, **square root**, and **cube root** functions in the app as well, but don't do any of the others yet. Do your results match your expectations? yes